

Dr. Ravi Yadav

Project Scientist III

Mausam Mission

Indian Institute of Tropical Meteorology, Pune

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Email: yadavravi38@gmail.com

ResearchGate: <https://www.researchgate.net/profile/Ravi-Yadav-10>

Google Scholar: <https://scholar.google.co.in/citations?user=5hE-ofEAAAAJ&hl=en>

CAREER OBJECTIVE

A good approach to understanding atmospheric composition and its chemistry using observational techniques and models

RESEARCH INTERESTS

Understanding the atmospheric chemical processes by using an outdoor simulation chamber (100 m³) and a small chamber as well

Atmospheric chemistry and trace gases, precursors of ozone in the troposphere

Air quality and meteorology

Long-range transport of trace constituents

Use of reanalysis data from CAMS

Trend analysis of long-term data to understand the air pollution climatology

Atmospheric chemical modeling using the Master Chemical Mechanism (MCM)

ACADEMIC QUALIFICATIONS

Ph.D. in Physics, Department of Physics, Mohanlal Sukhadia University (MLSU), Udaipur, India

June 03, 2015

Thesis entitled: *"Impact of Meteorological Parameters on Air Pollutants over Tropical Station Udaipur."*

Area of Research: Atmospheric Science-Physics

Coursework percentage: **First class with 67.66%**

Work done:

- ✓ In collaboration with the Indian Institute of Tropical Meteorology, Pune, under the "Modeling Atmospheric Pollutants and Networking-(MAPAN)" scheme.
- ✓ Most of my Ph.D. work has been done in collaboration with the Space and Atmospheric Science Division, Physical Research Laboratory, Ahmedabad.

M.Sc. in Physics, Mohanlal Sukhadia University, Udaipur, **April 2009**

Percentage: **First class with 67%**

Subjects: Solid state physics; Nuclear physics; Atomic spectroscopy; Quantum physics; Statistical physics; Astronomy and Astrophysics; Electronic Physics etc.

B.Sc. in Physics, University of Kota, **June 2006**

Percentage: **First class with 61%**

Subjects: Physics; Chemistry; Mathematics

RESEARCH APPOINTMENTS

Special Research Assistant/Post-Doctoral Fellow

25 October 2023 to 31 December 2025

Laboratory of Atmospheric Environment and Pollution Control (LAEPC), Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, Beijing 100085, China

- ✓ Understanding the formation mechanism of particulate sulfate under controlled real ambient air conditions
- ✓ Simulated role of NO_x on ozone formation from the oxidation of VOC under natural light in the 100 m³ Chamber

Research Fellow

19 September 2022 to 16 July 2023

NUS Environmental Research Institute, National University of Singapore, Singapore

- ✓ Study of roadside emissions, including VOCs and particles, in the context of the air quality
- ✓ Air Sampling at the heavy-traffic site

Project Scientist B

26 June 2019 to 13 September 2022

PROMOTE-INDO-UK Atmospheric Pollution and Human Health (APHH) project, Indian Institute of Tropical Meteorology, Pune, India

- ✓ Study of experimental-related measurements of trace gases and ozone precursors (O₃) in Indian megacities to study the air quality and their emission sources in cities.
- ✓ Understanding the chemistry of volatile organic compounds (VOCs) using PTR-MS
- ✓ The role of biogenic emissions and local meteorology on VOCs distribution

Visiting Researcher

01 August 2018 to 31 October 2018

Department of Engineering and Environmental Science, Fudan University, Shanghai, China

- ✓ Study of New particle Formation at a high-altitude site (Mt. Ailao ~ 3000m) of China.

Post-Doctoral Research Associate

01 January 2018 to 13 April 2018

School of Natural Sciences and Engineering, National Institute of Advanced Studies (NIAS), IISc Campus, Bangalore, India.

- ✓ Study particulate matter in coal mines and power plants using dispersion modeling.

Visiting Fellow

01 February 2017 to 30 November 2017

Department of Space and Atmospheric Sciences Division (SPA-SC), Physical Research Laboratory (PRL), India.

- ✓ Measurements of volatile organic compounds (VOCs) in the air using PTR-TOF-MS
- ✓ Field campaign: Air Sampling at different locations in India
- ✓ Analysis of VOCs using TD-GC-FID

Post-Doctoral Fellow

10 November 2014 to 09 January 2017

Department of Space and Atmospheric Sciences Division (SPA-SC), Physical Research Laboratory (PRL), India.

- ✓ Study of VOCs using proton transfer reaction mass spectrometry time of flight (PTR-TOF-MS) and (TD-GC-FID) system
- ✓ Simultaneous measurements of surface O₃, CO, and NO_x using online optical techniques (Thermo analyzers)
- ✓ Study the relations between VOC-CO-NO_x in context to the photochemical production of ozone at Ahmedabad

Research Scholar

11 February 2011 to 31 October 2014

Department of Physics, Mohanlal Sukhadia University, Udaipur, India.

I completed my doctorate under the project “Modeling Atmospheric Pollutants and Networking (MAPAN),” which was governed by the Indian Institute of Tropical Meteorology, Pune, Ministry of Earth Sciences (MoES), Govt. of India, on the university campus.

As a Research Scholar, I was engaged with the following type of work:

- ✓ Investigated the linkages of anthropogenic emissions and local meteorology in the rapid increase of particulate matter using three years of measurements at a foothill city in India.
- ✓ Investigated the dominance of combustion-related sources in the dry (winter-pre-monsoon) season.
- ✓ I have prepared a consistent data set of O₃, NO, NO₂, NO_x, CO, PM₁₀, and PM_{2.5}, from April 2010 to November 2014 by quality control check, which was used to understand the assessment of air quality.
- ✓ Based on the MAPAN (Modeling Atmospheric Pollutants and Networking) data, I have worked on air quality and mitigation strategies.
- ✓ Understanding of the Air Quality Index (AQI) using the long-term data (2010-2014) of O₃ CO and NO₂ for Udaipur, which showed that air quality was mostly in good ranges (except for some periods).
- ✓ I was one of the contributing authors of all journal publications and conference proceedings/workshops.

Project Fellow

01 April 2010 to 09 February 2011

- ✓ I have joined the MAPAN (Modeling of Atmospheric Pollutants and Networking) project, IITM-MoES, in setting up the ozone pollution laboratory in an urban site of Udaipur in Western India to investigate the variability of surface ozone (O₃), oxides of nitrogen (NO_x), carbon monoxide (CO), as well as particulate matters (PMs) within the lower troposphere.
- ✓ I was engaged in maintaining the air quality station through proper monitoring, modifications, development, and calibrations of various instruments (listed below) used for measuring atmospheric pollutants.
- ✓ Responsibilities to operate the automatic weather station (AWS)

TEACHING APPOINTMENT

Assistant Professor

02 January 2019- 31 May 2019

- ✓ Taught **Engineering Physics** for B. Tech students in the department of Physics, Poornima College of Engineering (PCE), Jaipur, Rajasthan, India.

Guest Lecturer

01 July 2009-30 March 2010

- ✓ Taught **Basic Physics** for B.Sc. students in the Department of Physics, Vidhya Bhawan Rural Institute, Udaipur, Rajasthan, India.

HONOURS & ACHIEVEMENTS/SCIENTIFIC RECOGNITION

- ✓ The Project "**Exploration of the Formation Mechanism of Particulate Matter Sulfate under controllable and near-actual Atmospheric Environmental Conditions**" has been awarded/funded by the **Beijing Natural Science Foundation** at the Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences.
Name: **Ravi Yadav**
Funding ID: **1S24098**
Amount: **27,460.00 USD**
- ✓ Guest Associate editor-2021 for the special theme "Anthropogenic Trace Gases and their Linkages to Meteorology and Climate Change" in Frontiers in Sustainable Cities, Climate Change, and Cities
- ✓ The **review article** "Observations of Trace Gases in the Earth's Lower Atmosphere: Instrumentation and Platform" in the 25 June 2020 issue of **Current Science** has been selected as a **Highlight**.
<https://www.currentscience.ac.in/Volumes/118/12/1893.pdf>
<https://www.currentscience.ac.in/Volumes/118/12/1862.pdf>
- ✓ "**Reviewing Certificates**" from Environmental Pollution, Atmospheric Research, Atmospheric Environment, etc. Journals.
- ✓ First prize in an Essay competition during Hindi Pakhwada Week 2019, held at the Indian Institute of Tropical Meteorology, Pune, an autonomous Institute of the Ministry of Earth's Sciences (MoES), Govt. of India.
- ✓ Early Career Scientist Travel Award by IGAC and WMO to present a paper at 14th International Global Atmospheric Chemistry (**IGAC**)-2016 Science Conference, Breckenridge, Colorado, USA.
- ✓ Post-Doctoral Fellowship, Department of Space, at Physical Research Laboratory, Ahmedabad (Gujarat), India.
- ✓ MAPAN Junior Research Fellow (JRF), Ministry of Earth Sciences, Indian Institute of Tropical Meteorology, Pune at Mohanlal Sukhadia University, Udaipur (Raj.) India.

COMPUTATIONAL SKILLS

- ✓ Operating system: Windows; Linux
- ✓ Plotting packages: Python, GrADS, Sigma plot, Origin, Matlab, and Igor with statistical analysis
- ✓ Scientific data formats: netCDF; HDF
- ✓ Positive Matrix Factorization (PMF) model for source apportionment
- ✓ Cluster analysis of trajectories
- ✓ Analysis of CAMS reanalysis datasets

INSTRUMENTATION SKILLS

Proficiency in installation, operation, maintenance, data quality control and quality assurance, Calibration, monitoring air quality and weather parameters, analysis of data from various types of scientific instruments i.e.

- ✓ Operating and managing the biggest Outdoor Simulation Chamber (100 m³), which is equipped with a range of in-situ and off-line analytical instrumentation
- ✓ Carbonyls compounds sampling using 2,4-dinitrophenylhydrazine (DNPH) silica column sampling tubes
- ✓ VOCs measurement-VOCUS-PTR-TOF (TOFWERK)
- ✓ VOCs measurement- PTR-TOF-MS₈₀₀₀ and PTR-QMS₅₀₀ (Ionicon Analytik GmbH Innsbruck, Austria)
- ✓ GCU (gas calibration Unit) for calibration purposes for PTR-MS
- ✓ NMVOCs analysis - GC-FID (7890A-G3440A, Agilent, Palo Alto, CA, USA) coupled with a TD system (UNITY2 with Air Server 3 channel module, Markes International, UK)
- ✓ UV photometric-based analyzer for Ozone (O₃), Chemiluminescence based technique for oxides of nitrogen (NO_x), IR-based technique for Carbon monoxide (CO), fluorescence technology based SO₂ analyser
- ✓ TSI scanning mobility particle sizer (SMPS)
- ✓ Beta ray attenuation technique for PM_{2.5} and PM₁₀ using BAM 1020-Met one Instrument
- ✓ APM550 MFC for sampling the fine particle and coarser particles
- ✓ Automatic weather sensors (AWS)
- ✓ Air Sampling of particles: Personal Cascade Impactor Sampler (PCIS) and Personal Environmental Monitor (PEM) using Zflour and quartz filter
- ✓ Xact® 625i Ambient Continuous Multi-Metals Monitor

REFEREED JOURNAL PUBLICATIONS

Published: 41

(Google Scholar: Total Citations-1658; h-index-22; i10-index-31) (Scopus: Total Citations-1412; h-index-22 based on 39 documents)

2025

1. Yang Jiao, Yangang Ren, Jinhe Wang, **Ravi Yadav**, Shan Chen, Chengtang Liu, Jiangshan Mu, Likun Xue, Yujing Mu, Abdelwahid Mellouki (2025), Influence of tourism on the local air quality in the Mountain Laoshan Forest scenic areas. *Atmospheric Environment*, 353, 121229.
<https://doi.org/10.1016/j.atmosenv.2025.121229>

(Citations- 1)

2. L. K. Sahu, Mansi Gupta, Nidhi Tripathi, **Ravi Yadav**, Tanzil Gaffar Malik, Mizuo Kajino (2025), Effect of Different Sources and Meteorological Processes on the Variability of VOC Composition in a Metropolitan City of Western India During Summer Season. *Journal of Geophysical Research-Atmospheres*, 130(9), e2024JD040867.

<https://doi.org/10.1029/2024JD040867>

(Citations-1)

3. Jinhe Wang, Ouyang Li, Pengcheng Zhang, Xue Yang, **Ravi Yadav**, Shan Chen, Yuhong Liu, Chao Zhu, Yangang Ren, Abdelwahid Mellouki (2024), Atmospheric chemistry of the coastal area is influenced by the convergence between the inland and marine air: Insight into carbonyl compounds. *Journal of Environmental Sciences*, 154, 859-870.

<https://doi.org/10.1016/j.jes.2024.06.019>

(Citations- 1)

4. Saroj Kumar Sahu, Ashirbad Mishra, Poonam Mangaraj, **Ravi Yadav**, Mahesh Chandra Sahu, Gufran Beig, Bhishma Tyagi, Sachin S Gunthe, B Anjan Kumar Prusty, Madhusmita Mishra (2025), Source-specific fine particulates emission linked to prevalence of ophthalmic cases in India. *Scientific Reports*, 15, 11183.

<https://doi.org/10.1038/s41598-024-82914-6>

(Citations- 0)

2024

5. Yangang Ren, Jinhe Wang, Wenya Niu, Yang Jiao, Chenglong Zhang, **Ravi Yadav**, Yonghong Wang, Junfeng Liu, Chunshan Liu, Yujing Mu, Hong He (2024), The performance of a 100 m³ outdoor atmospheric simulation chamber in China. *Science of the Total Environment*, 957, 177567.

<https://doi.org/10.1016/j.scitotenv.2024.177567>

(Citations- 3)

6. **Ravi Yadav**, V. Anand, S.K. Sahu, R.K. Kunchala, B. Tyagi, G. Beig (2024), Editorial: Anthropogenic trace gases and their linkages to meteorology and climate change. *Frontier in Sustainable Cities*, 6, 1379626.

<https://doi.org/10.3389/frsc.2024.1379626>

(Citations- 0)

2023

7. Saroj Kumar Sahu, Pallavi Sahoo, Poonam Mangaraj, Gufran Beig, Bhishma Tyagi, Basanta Samal, Ashirbad Mishra, **Ravi Yadav** (2023), Identification and Quantification of Emission Hotspots of Air Pollutants over Bhubaneswar: A Smart City in Eastern India. *Aerosol and Air Quality Research*, 23, 230049.

<https://doi.org/10.4209/aaqr.230049>

(Citations- 19)

8. R. Kalbande, B. Kumar, S. Maji, **Ravi Yadav**, K. Atey, D.S. Rathore, G. Beig (2023), Machine learning based quantification of VOC contribution in surface ozone. *Chemosphere*, 326, 138474.

<https://doi.org/10.1016/j.chemosphere.2023.138474>

(Citations- 26)

9. P. Kumar, G. Beig, S.K. Sahu, **Ravi Yadav**, S. Maji, V. Singh, B.R. Bamniya (2023), Development of a high-resolution emissions inventory of carbonaceous particulate matters and their growth during 2011–2018 over India. *Atmospheric Environment*, 303, 119750.
<https://doi.org/10.1016/j.atmosenv.2023.119750>

(Citations- 21)

2022

10. Ritesh Kalbande, **Ravi Yadav**, Sujit Maji, D.S. Rathore, G. Beig (2022), Characteristics of VOCs and their contribution to O₃ and SOA formation across seasons over a metropolitan region in India. *Atmospheric Pollution Research*, 13(8), 101515.
<https://doi.org/10.1016/j.apr.2022.101515>

(Citations- 21)

11. S.K. Sahu, P. Mangaraj, B. Tyagi, **Ravi Yadav**, Oscar Paul, S. Chaulya, C. Pradhan, N. Das, P. Sahoo, G. Beig (2022), Estimating NO_x, VOC, and CO variability over India's 1st smart city: Bhubaneswar. *Frontiers in Environmental Science*, 10, 997026.
<https://doi.org/10.3389/fenvs.2022.997026>

(Citations- 6)

12. **Ravi Yadav**, Gufran Beig, Vrinda Anand, Ritesh Kalbande, Sujit Maji (2022), Tracer-based characterization of source variations of ambient isoprene mixing ratios in a hilly megacity, India, influenced by the local meteorology. *Environmental Research*, 205, 112465. <https://doi.org/10.1016/j.envres.2021.112465>

(Citations- 16)

13. L.K. Sahu, Nidhi Tripathi, Mansi Gupta, Vikas Singh, **Ravi Yadav**, Kashyap Patel (2022), Impact of COVID-19 pandemic lockdown in ambient concentrations of aromatic volatile organic compounds in a metropolitan city of western India. *Journal of Geophysical Research-Atmospheres*, 127, e2022JD036628.
<https://doi.org/10.1029/2022JD036628>

(Citations- 37)

14. **Ravi Yadav**, Pushpendra Vyas, Praveen Kumar, Lokesh Kumar Sahu, Umang Kumar V Pandeya, Nidhi Tripathi, Mansi Gupta, Vikram Singh, Pragnesh N Dave, Devendra Singh Rathode, Gufran Beig, S.N.A. Jaaffrey (2022), Particulate-matter pollution in urban cities of India during unusually restricted anthropogenic activities. *Frontiers in Sustainable Cities*, 4, 792507.
<https://doi.org/10.3389/frsc.2022.792507>

(Citations- 13)

15. Poonam Mangaraj, Saroj Kumar Sahu, Gufran Beig, **Ravi Yadav** (2022), A comprehensive high-resolution gridded emission inventory of anthropogenic sources of air pollutants in Indian megacity, Kolkata. *SN Applied Sciences*, 4, 117.
<https://doi.org/10.1007/s42452-022-05001-3>

(Citations- 40)

16. Suvarna Tikle, Abhilash Panicker, Aditi Rathod, **Ravi Yadav**, Gufran Beig (2022), Assessment of air quality index in a most polluted megacity of Asia. *International Journal of Engineering Applied Sciences and Technology*, 6, 15-21.
<https://doi.org/10.33564/IJEAST.2022.v06i11.004>

(Citations- 1)

2021

17. Sokhi, R., et al. (**Ravi Yadav**, one of the authors) (2021), A global observational analysis to understand changes in air quality during exceptionally low anthropogenic emission conditions. *Environmental International*, 157, 106818, <https://doi.org/10.1016/j.envint.2021.106818>

(Citations- 232)

18. S. Maji, **Ravi Yadav**, G. Beig, S.S. Gunthe, N. Ojha (2021), On the processes governing the variability of PTR-MS based VOCs and OVOCs in different seasons of a year over hillcity megacity of India. *Atmospheric Research*, 261, 105736, <https://doi.org/10.1016/j.atmosres.2021.105736>

(Citations- 14)

19. Nidhi Tripathi, L.K. Sahu, K. Patel, A. Kumar, **Ravi Yadav** (2021), Ambient air characteristics of biogenic volatile organic compounds at a tropical evergreen forest site in central Western Ghats of India. *Journal of Atmospheric Chemistry*, 78, 139-149. <https://doi.org/10.1007/s10874-021-09415-y>

(Citations- 16)

2020

20. Nidhi Tripathi, L.K. Sahu, Arvind Singh, **Ravi Yadav**, A. Patel, K. Patel, P. Meenu (2020), Elevated levels of biogenic non-methane hydrocarbons in the marine boundary layer of the Arabian Sea during the inter-monsoon. *Journal of Geophysical Research-Atmospheres*, 125, e2020JD032869.

<https://doi.org/10.1029/2020JD032869>

(Citations- 31)

21. **Ravi Yadav**, N. Korhale, V. Anand, A. Rathod, S. Bano, R. Shinde, R. Latha, S.K. Sahu, B.S. Murthy, Gufran Beig (2020), COVID-19 lockdown and air quality of SAFAR-India metro cities. *Urban Climate*, 34, 100729.

<https://doi.org/10.1016/j.uclim.2020.100729>

(Citations- 51)

22. Gufran Beig, S. Bano, S.K. Sahu, V. Anand, N. Korhale, A. Rathod, **Ravi Yadav**, P. Mangaraj, B. S. Murthy, S. Singh, R. Latha, R. Shinde (2020), COVID-19 and environmental-weather markers: Unfolding baseline levels and veracity of linkages in tropical India. *Environmental Research*, 191, 110121.

<https://doi.org/10.1016/j.envres.2020.110121>

(Citations- 36)

23. Pragnesh Dave, L.K. Sahu, Nidhi Tripathi, Samiksha Bajaj, **Ravi Yadav**, Kashyap Patel (2020), Emissions of non-methane volatile organic compounds from a landfill site in a major city of India: impact on local air quality. *Heliyon*, 6, e04537, <https://doi.org/10.1016/j.heliyon.2020.e04537>

(Citations- 31)

24. L.K. Sahu, **Ravi Yadav**, Nidhi Tripathi (2020), Aromatic compounds in a semi-urban site of western India: Seasonal variability and emission ratios. *Atmospheric Research*, 246, 105114.

<https://doi.org/10.1016/j.atmosres.2020.105114>

(Citations- 46)

25. L.K. Sahu, Nidhi Tripathi, **Ravi Yadav** (2020), Observations of trace gases in the earth's lower atmosphere: instrumentation and platform. *Current Science*, 118, 1893-1902. <https://doi.org/10.18520/cs/v118/i12/1893-1902>.

(Citations- 13)

26. Nidhi Tripathi, L.K. Sahu, Arvind Singh, **Ravi Yadav**, K.K. Karati (2020), High levels of Isoprene in the Marine Boundary Layer of the Arabian Sea during Spring Inter-Monsoon: Role of Phytoplankton Blooms. *ACS Earth and Space Chemistry*, 4(4), 583-590. <https://doi.org/10.1021/acsearthspacechem.9b00325>.

(Citations- 23)

27. Sujit Maji, G. Beig, **Ravi Yadav** (2020), Winter VOCs and OVOCs measured with PTR-MS at an urban site of India: Role of emissions, meteorology and photochemical sources. *Environmental Pollution*, 258, 113651. <https://doi.org/10.1016/j.envpol.2019.113651>

(Citations- 72)

2019

28. **Ravi Yadav**, L.K. Sahu, Nidhi Tripathi, D. Pal, G. Beig, S.N.A. Jaaffrey (2019), Investigation of emission characteristics of NMVOCs over urban site of Western India. *Environmental Pollution*, 252, 245-255. <https://doi.org/10.1016/j.envpol.2019.05.089>

(Citations- 66)

29. **Ravi Yadav**, L.K. Sahu, G. Beig, Nidhi Tripathi, Sujit Maji, S.N.A. Jaaffrey (2019), The role of local meteorology on ambient particulate and gaseous species at an urban site of western India. *Urban Climate*, 28, 100449. <https://doi.org/10.1016/j.uclim.2019.01.003>

(Citations- 28)

2018

30. L.K. Sahu, Nidhi Tripathi, Varun Sheel, Mizuo Kajino, Makoto Deushi, **Ravi Yadav**, Philippe Nedelec (2018), Impact of the tropical cyclone Nilam on the vertical distribution of carbon monoxide over Chennai on the Indian peninsula. *Quarterly Journal of the Royal Meteorological Society*, 144, 1091-1105. <https://doi.org/10.1002/qj.3276>

(Citations-7)

2017

31. **Ravi Yadav**, L.K. Sahu, G. Beig, Nidhi Tripathi, S.N.A. Jaaffrey (2017), Ambient particulate matter and carbon monoxide at an urban site of India: Influence of anthropogenic emissions and dust storms. *Environmental Pollution*, 225, 291-303. doi.org/10.1016/j.envpol.2017.01.038

(Citations- 86)

32. L.K. Sahu, Nidhi Tripathi, **Ravi Yadav** (2017), Contribution of biogenic and photochemical sources to ambient VOCs during winter-summer transition at a semi-arid urban site in India. *Environmental Pollution*, 229, 595-606. <https://doi.org/10.1016/j.envpol.2017.06.091>

(Citations- 78)

33. L.K. Sahu, Varun Sheel, M Kajino, M Deushi, Sachin S Gunthe, P.R. Sinha, **Ravi Yadav**, Devendra Pal, P. Nedelec, Valérie Thouret, Herman G Smit (2017), Impact of tropical convection and ENSO variability in vertical distributions of CO and O₃ over an urban site of India. *Climate Dynamics*, 49, 449-469.

<https://doi.org/10.1007/s00382-016-3353-7>

(Citations- 14)

2016

34. **Ravi Yadav**, L.K. Sahu, G. Beig, S.N.A. Jaaffrey (2016), Role of long-range transport and local meteorology Variability in seasonal variation of surface ozone and its precursors at an urban site in India. *Atmospheric Research*, (176-177), 96-107,

<https://doi.org/10.1016/j.atmosres.2016.02.018>

(Citations- 130)

35. L.K. Sahu, **Ravi Yadav**, D. Pal (2016), Source identification of VOCs at an urban site of western India: Effect of marathon events and anthropogenic emissions. *Journal of Geophysical Research-Atmosphere*, 121, 2416-2433,

<https://doi.org/10.1002/2015JD024454>

(Citations- 116)

36. L.K. Sahu, D. Pal, **Ravi Yadav**, J. Munkhtur (2016), Aromatic VOCs at major road junctions of a Metropolis in India: Measurements using TD-GC-FID and PTR-TOF-MS instruments, *Aerosol Air Quality Research*, 17, 2405-2420.

<https://doi.org/10.4209/aaqr.2015.11.0643>

(Citations- 41)

2015

37. L.K. Sahu, Varun Sheel, Kumud Pandey, **Ravi Yadav**, P Saxena, S Gunthe (2015), Regional biomass burning trends in India: Analysis of satellite fire data. *Journal of Earth System Science*, 124, 1377-1387.

<https://doi.org/10.1007/s12040-015-0616-3>

(Citations- 96)

2014

38. **Ravi Yadav**, G. Beig, S.N.A. Jaaffrey (2014), The linkages of anthropogenic emissions and meteorology in the rapid increase of particulate matter at a foothill city in the Aravali range of India. *Atmospheric Environment*, 85, 147-151.

<https://doi.org/10.1016/j.atmosenv.2013.09.007>

(Citations- 43)

39. **Ravi Yadav**, L.K. Sahu, S.N.A. Jaaffrey, G. Beig (2014), Temporal Variation of Particulate Matter (PM) and potential sources at an urban site of Udaipur in Western India. *Aerosol Air Quality Research*, 14, 1613-1629.

<https://doi.org/10.4209/aaqr.2013.10.0310>

(Citations- 118)

40. **Ravi Yadav**, L.K. Sahu, S.N.A. Jaaffrey, G. Beig (2014), Distribution of Ozone and related trace gases at a semi-urban site of Udaipur in Western India. *Journal of Atmospheric Chemistry*, 71, 125-144.

<https://doi.org/10.1007/s10874-014-9286-9>

(Citations- 53)

41. D. Pareek, S.N.A. Jaaffrey, K.P. Talesra, **Ravi Yadav**, S. Ameta (2013). Experimental study of Variation of Secondary Cosmic Gamma Ray Flux and Energy during Partial Solar Eclipse of 4th January 2011 at Udaipur, India. *Research Journal of Physical Sciences*, 1(5), 22-30.
(Citations- 07)

BOOK CHAPTERS (PEER-REVIEWED)

Published: 01

2021

42. **Ravi Yadav**, Pujal Trivedi, L.K. Sahu, G. Beig, Nidhi Tripathi (2021), Air Pollution Modeling. **Air Pollution and Environmental Health: Environmental Chemistry for a Sustainable World**, Springer Nature Singapore, 20, 37-55.
https://doi.org/10.1007/978-981-15-3481-2_3.
(Citations- 3)

PEER-REVIEWED PROCEEDINGS

Published: 2

2012

1. **Ravi Yadav**, L.K. Sahu, S.N.A. Jaaffrey, G. Beig (2012), Distribution of Particulate Matter, CO and NO_x at a semi-urban site of Udaipur in Rajasthan India. **Indian Aerosol Science and Technology Association (IASTA) Bulletin**, 20 (1-2), ISSN 0971-4570, 49-51.

2014

2. **Ravi Yadav**, L.K. Sahu, S.N.A. Jaaffrey, G. Beig (2014), Correlation studies of NMHCs and other trace gases at a tropical semi urban site Udaipur in Western India: A MAPAN based study, International Tropical Meteorology Symposium (**INTROMET-2014**), 130.

PAPER PRESENTATION/PARTICIPATION IN CONFERENCES/SYMPOSIUMS

Total number of presentations: 19

2025

1. **Ravi Yadav**, Wang Yue, Yangang Ren, Abdelwahid Mellouki, Yujing Mu, (2025), Sulfate formation from the oxidation of SO₂ by NO₂. **Global Youth Conference-International Youth Forum on Green and Low-Carbon Development**, May 11-16, 2025, Qingdao.

2024

2. **Ravi Yadav**, Yangang Ren, Abdelwahid Mellouki, Yujing Mu, Understanding the formation mechanism of particulate sulfate under controlled real ambient air. **16th International Commission on Atmospheric Chemistry and Global Pollution (iCACGP)-18th International Global Atmospheric Chemistry (IGAC)**, September 09-13, 2024, Kuala Lumpur, Malaysia.

2021

3. **Ravi Yadav** and Gufran Beig, Biogenic and anthropogenic isoprene (C₅H₈) in the urban atmosphere. **WMO-GAW Symposium-2021**, June 28-2 July 2021.

2020

4. Participated in virtual “NERC/MoES Process analysis, observations and modelling - Integrated solutions for cleaner air for Delhi (PROMOTE) – SCIENCE MEETING” under the theme “**Air quality in Delhi**” organised by **Analysis of NERC/MoES**, December 08-09, 2020.

5. Participated in a virtual workshop under the theme “Changes in atmospheric composition during the Covid-19 lockdowns” organised by **Analysis of eMissions using Observations (AMIGO)/IGAC**, November 03, 2020.

2019

6. **Ravi Yadav**, L.K. Sahu, G. Beig, Nidhi Tripathi, S. N. A. Jaaffrey, Features of ambient particulate matters (PMs) and carbon monoxide (CO) at an urban atmosphere in India: Impact of meteorology and long-range transport. **National Space Science Symposium (NSSS)-2019**, January 29-31, 2019, University of Pune, Pune.

2017

7. Nidhi Tripathi, L.K. Sahu, **Ravi Yadav**, Mono-terpenes in urban sites of India: A precursors of biogenic SOA. **International Conference on Aerosol Climate Change Connection (AC3)**, April, 25-27, 2017, Bose Institute, Darjeeling.
8. Nidhi Tripathi, L. K. Sahu, **Ravi Yadav**, Measurement of VOCs in an urban site of India: Application of PTR-TOF-MS instruments. **European Research Course on Atmosphere (ERCA-2017)**, January 2017, Grenoble, France.

2016

9. Participated in Indian Aerosol Science and Technology Association (IASTA-2016) conference under the theme “**Aerosol and Climate change: Insights and Challenges**” organised by Physical Research Laboratory, Ahmedabad, December 06-08, 2016.
10. **Ravi Yadav**, D. Pal, L.K. Sahu, Emission ratios of C₂-C₇ non-methane hydrocarbons at an urban site of western India. **IGAC science conference-2016**, September 25-29, 2016, Breckenridge, Colorado, USA.
11. L.K. Sahu, **Ravi Yadav**, and D. Pal, Measurements of VOCs using PTR-MS-TOF system: Impact of anthropogenic and natural sources in ambient air. **National space science symposium-2016**, February 9-12, 2016, Space Physics Laboratory, VSSC, India.

2015

12. **Seminar entitled** “Study of ozone and precursors gases at an urban site of Udaipur, India.” K.R. Ramanathan Auditorium on September 29, 2015, **Physical Research Laboratory (PRL), India**.
13. **Ravi Yadav**, L.K. Sahu, S. N. A. Jaaffrey, G. Beig, Variability of surface O₃, CO and NO_x at an urban site of western India: Role of long-range transport and meteorology. **IC-CAST, Indian JSPS Alumni Association**, August 7-8, 2015, Banaras Hindu University, Varanasi, India.

2014

14. **Ravi Yadav**, L.K. Sahu, S. N. A. Jaaffrey, G. Beig, Correlation Studies of NMHCs and Other Trace Gases at a Tropical Semi Urban Site Udaipur in Western India: A MAPAN based study. **International Tropical Meteorology Symposium (INTROMET-2014)**, Jointly Organized by the Indian Meteorological Society and SRM University, February 21-24, 2014, SRM University, Chennai, India.

2013

15. **Ravi Yadav**, S. N. A. Jaaffrey, G. Beig, The linkages of anthropogenic emissions and meteorology in the rapid increase of particulate matter at a foothill city in the Aravali range

of India. **IITM-WMO Seminar (SAFAR)**, April 30, 2013, Indian Institute of Tropical Meteorology, Pune, India.

16. **Ravi Yadav**, L.K. Sahu, S. N. A. Jaaffrey, G. Beig, Observational Studies of Ozone and its Precursors at Udaipur in India. **International workshop on changing chemistry in changing climate: Monsoon (C4)**, May 1-3, 2013, Indian Institute of Tropical Meteorology, Pune, India.

2012

17. **Ravi Yadav**, L.K. Sahu, S. N. A. Jaaffrey, G. Beig, Seasonal and Diurnal Variations of Surface Ozone and its Precursors at Lake City, Udaipur, India. **Indo-German Workshop on Challenges and Opportunities in Air Pollution and Climate Change (CHOP-C)**, January 16-18, 2012, Indian Institute of Tropical Meteorology, Pune, India.
18. **Ravi Yadav**, L.K. Sahu, S. N. A. Jaaffrey, G. Beig, Temporal Variability of Surface Ozone and its Precursors along with Meteorological Parameters at a Tropical Station, Udaipur, India. **Tropmet-2012**, Organized by Indian Meteorological Society, November 20-22, 2012, Indian Institute of Remote Sensing (IIRS), Dehradun, India.
19. **Ravi Yadav**, L.K. Sahu, S. N. A. Jaaffrey, G. Beig, Distribution of Particulate Matter (PM), CO, and NO_x at a semi-urban site of Udaipur in Rajasthan, India. **Conference of Indian Aerosol Science and Technology Association (IASTA-2012)**, December 11-13, 2012, BARC, Navi Mumbai, India.

TRAINING ACTIVITIES

Total number of presentations: 03

2025

1. Participated in '**The fifth Sino-European School on Atmospheric Chemistry (SESAC5)**' at Fudan University, Shanghai, China, 05-18 November 2025.

2017

2. Participated in "**National training workshop on GHG measurement**" at the Indian Institute of Tropical Meteorology, Pune, from 10-14 July 2017.

2016

3. Participated in "**Aerosol Winter School 2016**" at the Indian Institute of Technology, Madras, from 16-19 February 2016.

INVITED TALKS

2021

1. "**Air Pollution: Role of COVID-19**" organized by Department of Physics, R. N. T. College, Kapasan, Chittorgarh, Rajasthan, April 03, 2021.
2. "**Emission of air pollutants and their roles in the earth's atmosphere**," organized by HRDC, Utkal University for refresher courses in environmental studies, March 19, 2021.
3. "**Our atmosphere, air pollution, and ozone hole**" organized by the IITM-ENVIS center on National Science Day-2021, February 26, 2021.

2020

4. "**Trace gases in the earth's atmosphere**" organized by the Department of Physics, Sangam University, December 24, 2020.

5. **“The atmosphere and air pollution”** organized by the Department of Physics, Pacific University, Udaipur, Rajasthan, June 10, 2020.

2019

6. **“Tropospheric ozone and related trace gases in the atmosphere,”** organized by the Department of Physics, R. N. T. College, Kapasan, Chittorgarh, Rajasthan, March 03, 2019.

PERSONAL DETAIL

Date of birth: 03 August 1986

Sex: Male

Marital status: Married

Nationality: Indian

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CURRENT ADDRESS

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PERSONAL QUALITIES

I am a hardworking and enthusiastic learner. I have good Interpersonal and managerial skills. I can familiarise myself and work successfully in any atmosphere.